



MATHEMATICS (EXTENSION 2)

2013 HSC Course Assessment Task 1

November 30, 2012

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided on page 3.

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Fletcher

☐ 12M4B – Mr Lam

☐ 12M4C – Ms Ziazaris

Marker's use only.

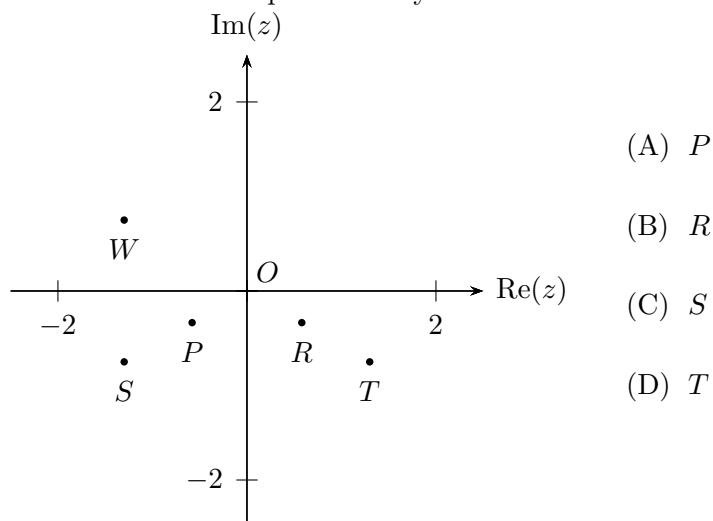
QUESTION	1 – 5	6	7	8	Total	%
MARKS	5	15	15	15	50	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. The point W on the Argand diagram below represents a number w where $|w| = 1.5$.
The number w^{-1} is best represented by **1**



2. Which of the following gives the value of z if $z^2 = 4 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$? **1**

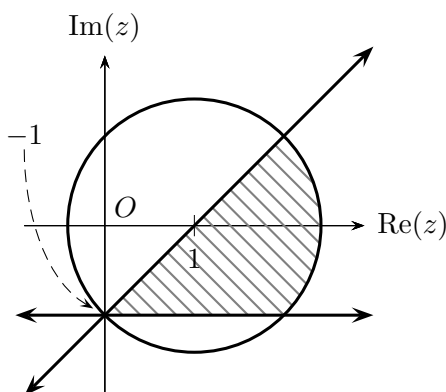
- (A) $\sqrt{3} + i$ or $-\sqrt{3} - i$ (C) $\sqrt{3} - i$ or $\sqrt{3} + i$
- (B) $1 - \sqrt{3}i$ or $-1 + \sqrt{3}i$ (D) $1 - \sqrt{3}i$ or $1 + \sqrt{3}i$

3. Let $z = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}$. What is the imaginary part of $z - i$? **1**

- (A) $-\frac{i}{2}$ (B) $-\frac{3i}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

4. What inequality given could define the shaded area?

1



(A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$.

(B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$.

(C) $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z - i) \leq \frac{\pi}{4}$.

(D) $|z - 1| \leq 1$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$.

5. Which of the following statement(s) is false, given $z = a + ib$ where $a \neq 0$ and $b \neq 0$?

1

(A) $z - \bar{z} = 2bi$

(C) $|z| + |\bar{z}| = |z + \bar{z}|$

(B) $|z|^2 = |z| |\bar{z}|$

(D) $\text{Arg}(z) + \text{Arg}(\bar{z}) = 0$

Answer grid for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

- | | | | | |
|-----|-----|-----|-----|-----|
| 1 – | (A) | (B) | (C) | (D) |
| 2 – | (A) | (B) | (C) | (D) |
| 3 – | (A) | (B) | (C) | (D) |
| 4 – | (A) | (B) | (C) | (D) |
| 5 – | (A) | (B) | (C) | (D) |

End of Section I.
Examination continues overleaf.

Section II: Short answer

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 6 (15 Marks)	Commence a NEW page.	Marks
(a) Find the exact value of a and b if $\frac{4+3i}{1+\sqrt{2}i} = a+ib$ such that $a, b \in \mathbb{R}$.		2
(b) i. Express $z = 1 + i\sqrt{3}$ in modulus-argument form.		2
ii. Hence show that $z^{10} + 512z = 0$.		2
(c) Given $z_1 = 4\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right)$ and $z_2 = 2\left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12}\right)$		
i. On an Argand diagram, draw the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ representing z_1 , z_2 and $z_1 + z_2$ respectively.		2
ii. Hence or otherwise, find $ z_1 + z_2 $ in simplest exact form.		2
(d) i. Solve $z^4 + 1 = 0$, giving your answers in modulus-argument form.		3
ii. Plot these solutions on the Argand diagram.		1
iii. Find the exact area of the quadrilateral that they form.		1

Question 7 (15 Marks)

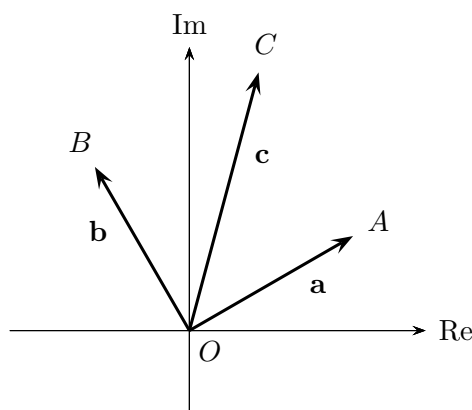
Commence a NEW page.

Marks

- (a) i. If $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$. **1**
- ii. Given that $z + \frac{1}{z} = \sqrt{2}$, find the value of $z^{10} + \frac{1}{z^{10}}$. **2**
- (b) Given that 1, ω and ω^2 are the cube roots of unity, find the value of $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega)$. **3**
- (c) i. Solve $w^2 = -11 - 60i$ for w , writing your answer in the form $w = x + iy$ where $x, y \in \mathbb{R}$. **2**
- ii. Hence or otherwise, solve the equation **3**

$$z^2 - (1 + 4i)z - (1 - 17i) = 0$$

- (d) In the Argand diagram below, vectors **a**, **b**, **c** represent the complex numbers z_1 , z_2 and $z_1 + z_2$ respectively, where $z_1 = \cos \theta + i \sin \theta$ and $z_1 + z_2 = (1 + i)z_1$.



- i. Express z_2 in terms of z_1 , and show that $OACB$ is a square. **2**
- ii. Show that $(z_1 + z_2) \overline{(z_1 - z_2)} = 2i$. **2**

Question 8 (15 Marks)		Commence a NEW page.	Marks
(a)	i.	On the same diagram, sketch the locus of both $ z - 2 = 2$ and $ z = z - 4i $.	2
	ii.	What is the complex number represented by the point of intersection of these two loci?	1
(b)	i.	On an Argand diagram, sketch the locus of the point P representing z such that	2
		$\left z - (\sqrt{3} + i) \right = 1$	
	ii.	Find the set of possible values of $ z $ and the set of possible values for $\text{Arg } z$.	2
(c)	Let $z = x + iy$, where $x, y \in \mathbb{R}$ be the complex number satisfying the inequality		4
	$z\bar{z} + (1 - 2i)z + (1 + 2i)\bar{z} \leq 4$		
	Sketch the locus of z on an Argand diagram.		
(d)	$\arg(z - 2) = \arg(z + 2) + \frac{\pi}{4}$ is the locus of the point P representing z on an Argand diagram.		
	i.	Show with a diagram why this locus is an arc of a circle.	2
	ii.	Find the centre and radius of this circle.	2

End of paper.

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Suggested Solutions

Section I

1. (A) 2. (B) 3. (C) 4. (B) 5. (C)

Section II

Question 6 (Ziazaris)

(a) (2 marks)

$$\begin{aligned}
 & \frac{4+3i}{1+\sqrt{2}i} \times \frac{1-\sqrt{2}i}{1-\sqrt{2}i} \\
 &= \frac{4+3\sqrt{2}+i(3-4\sqrt{2})}{1+2} \\
 &= \frac{4+3\sqrt{2}}{3} + i \left(\frac{3-4\sqrt{2}}{3} \right) \\
 a &= \frac{4+3\sqrt{2}}{3} \quad b = \left(\frac{3-4\sqrt{2}}{3} \right)
 \end{aligned}$$

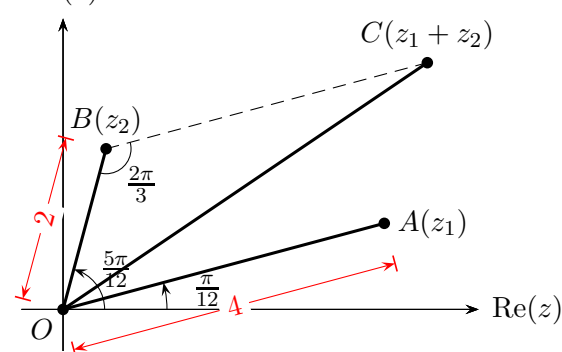
(b) i. (2 marks)

$$z = 1 + i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

ii. (2 marks)

$$\begin{aligned}
 z^{10} &= 2^{10} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)^{10} \\
 &= 2^{10} \left(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3} \right) \\
 &= 2^{10} \left(\cos \frac{-2\pi}{3} + i \sin \frac{-2\pi}{3} \right) \\
 512z &= 512 \times 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\
 &= 2^{10} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\
 \therefore z^{10} + 512z &= 2^{10} \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} + \frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \\
 &= 0
 \end{aligned}$$

(c) i. (2 marks)

ii. (2 marks) In $OACB$,

$$\angle AOB = \frac{4\pi}{12} = \frac{\pi}{3}$$

As $OACB$ is a parallelogram, then $\angle OBC = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Applying the cosine rule in $\triangle OBC$,

$$\begin{aligned}
 OC^2 &= |z_1 + z_2| \\
 &= 2^2 + 4^2 - 2(2)(4) \cos \frac{2\pi}{3} \\
 &= 20 - 8 \times \left(-\frac{1}{2} \right) = 28 \\
 \therefore OC &= |z_1 + z_2| = \sqrt{28} = 2\sqrt{7}
 \end{aligned}$$

(d) i. (3 marks)

$$\begin{aligned}
 z^4 + 1 &= 0 \\
 z^4 &= -1
 \end{aligned}$$

Letting $z = \cos \theta + i \sin \theta$,

$$\begin{aligned}
 & (\cos \theta + i \sin \theta)^4 \\
 &= \cos(\pi + 2k\pi) + i \sin(\pi + 2k\pi)
 \end{aligned}$$

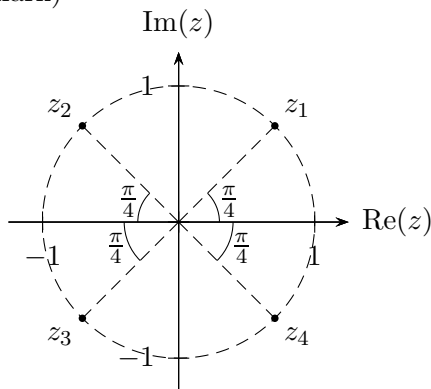
By De Moivre's Theorem,

$$\begin{aligned}
 \therefore \cos \theta + i \sin \theta &= \cos \left(\frac{\pi + 2k\pi}{4} \right) + i \sin \left(\frac{\pi + 2k\pi}{4} \right)
 \end{aligned}$$

where $k = 0, 1, 2, 3$

$$\begin{aligned}
 z_1 &= \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} & k=0 \\
 z_2 &= \cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) & k=1 \\
 z_3 &= \cos \left(\frac{-3\pi}{4} \right) + i \sin \left(\frac{-3\pi}{4} \right) & k=2 \\
 z_4 &= \cos \left(\frac{-\pi}{4} \right) + i \sin \left(\frac{-\pi}{4} \right) & k=3
 \end{aligned}$$

ii. (1 mark)



iii. (1 mark) Shape formed is a rhombus, with diagonals of length 2

$$A = \frac{1}{2}xy = \frac{1}{2} \times 2 \times 2 = 2$$

Question 7 (Fletcher)

(a) i. (1 mark)

$$\begin{aligned} z &= \cos \theta + i \sin \theta \\ \therefore z^n &= \cos n\theta + i \sin n\theta \\ z^{-n} &= \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta - i \sin n\theta \\ \therefore z^n + z^{-n} &= 2 \cos(n\theta) \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} z + \frac{1}{z} &= \sqrt{2} = 2 \cos \theta \\ \therefore \cos \theta &= \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \\ \therefore \theta &= \frac{\pi}{4} \end{aligned}$$

Now,

$$\begin{aligned} z^{10} + \frac{1}{z^{10}} &= 2 \cos(10\theta) \\ \therefore 2 \cos\left(10 \times \frac{\pi}{4}\right) &= 2 \cos\left(\frac{5\pi}{2}\right) \\ &= 2 \cos \frac{\pi}{2} = 0 \end{aligned}$$

(b) (3 marks)

$1, \omega, \omega^2$ are cube roots of unity, i.e.

$$\begin{aligned} \omega^3 - 1 &= 0 \\ (\omega - 1)(\omega^2 + \omega + 1) &= 0 \\ \therefore \omega^2 &= -\omega - 1 \end{aligned}$$

Also, $\omega^3 = 1$. Use these properties to reduce higher powers of ω to lower powers:

$$\begin{aligned} &(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) \\ &= (1 + \cancel{\omega} + \omega^2 + \omega + 2\omega^2) \\ &\quad (\cancel{1} + \cancel{\omega} + \omega^2 + \omega^2 + 2\omega) \\ &= (2\omega^2 + \omega)(\omega^2 + 2\omega) \\ &= \omega^2(2\omega + 1)(\omega + 2) \\ &= \omega^2(2\omega^2 + 5\omega + 2) \\ &= 2\omega^4 + 5\omega^3 + 2\omega^2 \\ &= 2\omega + 5 + 2\omega^2 \\ &= 2\omega + 5 + 2(-\omega - 1) \\ &= 5 - 2 = 3 \end{aligned}$$

(c) i. (2 marks)

$$\omega^2 = -11 - 60i$$

Let $\omega = x + iy$

$$\begin{aligned} \therefore (x + iy)^2 &= -11 - 60i \\ x^2 - y^2 + i(2xy) &= -11 - 60i \end{aligned}$$

Equating real and imaginary parts,

$$\begin{cases} x^2 - y^2 = -11 & (1) \\ 2xy = -60 & (2) \end{cases}$$

From (2),

$$\begin{aligned} xy &= -30 \\ y &= -\frac{30}{x} \\ \therefore x^2 - \left(\frac{-30}{x}\right)^2 &= -11 \\ x^2 - \frac{900}{x^2} &= -11 \\ x^4 - 900 &= -11x^2 \\ x^4 + 11x^2 - 900 &= 0 \end{aligned}$$

Letting $u = x^2$,

$$\begin{aligned} u^2 + 11u - 900 &= 0 \\ (u + 36)(u - 25) &= 0 \Rightarrow u = -36, 25 \\ \therefore x^2 &= 25 \\ x &= \pm 5 \quad y = \mp 6 \\ \therefore \omega &= \pm(5 - 6i) \end{aligned}$$

ii. (3 marks)

$$\begin{aligned}
 z^2 - (1 + 4i)z - (1 - 17i) &= 0 \\
 z &= \frac{(1 + 4i) \pm \sqrt{(1 + 4i)^2 + 4(1 - 17i)}}{2} \\
 &= \frac{(1 + 4i) \pm \sqrt{1 + 8i + 16i^2 + 4(1 - 17i)}}{2} \\
 &= \frac{(1 + 4i) \pm \sqrt{-15 + 8i + 4 - 68i}}{2} \\
 &= \frac{(1 + 4i) \pm \sqrt{-11 - 60i}}{2} \\
 &= \frac{(1 + 4i) \pm (5 - 6i)}{2}
 \end{aligned}$$

Positive solution:

$$\begin{aligned}
 z &= \frac{1 + 4i + 5 - 6i}{2} \\
 &= \frac{6 - 2i}{2} = 3 - i
 \end{aligned}$$

Negative solution:

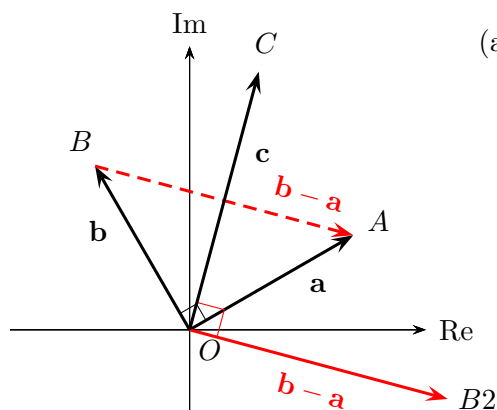
$$\begin{aligned}
 z &= \frac{1 + 4i - (5 - 6i)}{2} \\
 &= \frac{-4 + 10i}{2} = -2 + 5i
 \end{aligned}$$

(d) i. (2 marks)

$$\begin{aligned}
 z_1 + z_2 &= (1 + i)z_1 \\
 z_1 + z_2 &= z_1 + iz_1 \\
 \therefore z_2 &= iz_1
 \end{aligned}$$

- Hence z_2 is a rotation of z_1 by 90° , i.e. $\angle AOB = \frac{\pi}{2}$.
- As $OA = OB$ and $\angle AOB = 90^\circ$, $OACB$ is a square.

ii. (2 marks)



- As c is the diagonal of a square, then C will be representing the complex number $\sqrt{2}(\cos \theta + i \sin \theta)$.
- Also, $b - a$ is also the diagonal of the square $OACB$, it will also have modulus $\sqrt{2}$. In addition, it is a rotation of -90° of the vector c , hence $b - a$ (representing the complex number $z_1 - z_2$) is

$$\sqrt{2} \left(\cos \left(\theta - \frac{\pi}{2} \right) + i \sin \left(\theta - \frac{\pi}{2} \right) \right)$$

- $\overline{z_1 - z_2}$ is therefore

$$\begin{aligned}
 &\sqrt{2} \left(\cos \left(\theta - \frac{\pi}{2} \right) - i \sin \left(\theta - \frac{\pi}{2} \right) \right) \\
 &= \sqrt{2} \left(\cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right)
 \end{aligned}$$

- Multiplying $z_1 + z_2$ with $\overline{z_1 - z_2}$,

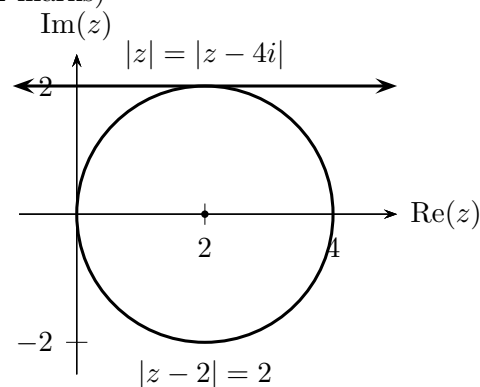
$$\begin{aligned}
 &\sqrt{2}(\cos \theta + i \sin \theta) \\
 &\times \sqrt{2} \left(\cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right) \\
 &= 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \\
 &= 2i
 \end{aligned}$$

Alternatively, expand via algebraic method:

$$\begin{aligned}
 z_2 &= iz_1 \\
 (z_1 + z_2)(\overline{z_1 - z_2}) &= (z_1 + iz_1)(\overline{z_1 - iz_1}) \\
 &= z_1(1 + i)\overline{z_1}(1 - i) \\
 &= z_1\overline{z_1}(1 + i)^2 \\
 &= \underbrace{|z_1|^2}_{=1} \times 2i = 2i
 \end{aligned}$$

Question 8 (Lam)

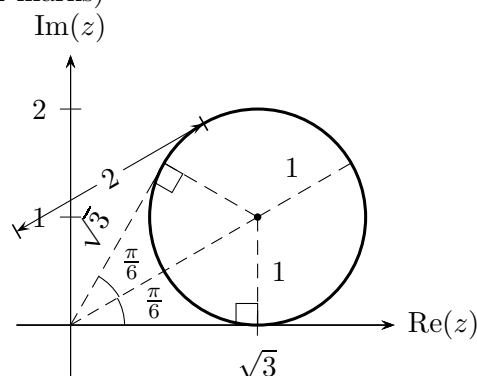
(a) i. (2 marks)



ii. (1 mark)

$$2 + 2i$$

(b) i. (2 marks)



ii. (2 marks)

$$0 \leq \text{Arg } z \leq \frac{\pi}{3} \quad 1 \leq |z| \leq 3$$

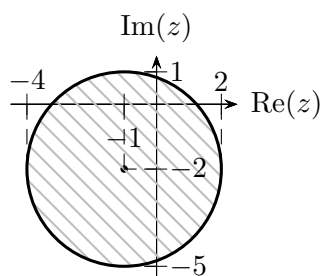
(c) (4 marks)

$$z\bar{z} + (1 - 2i)z + (1 + 2i)\bar{z} \leq 4$$

Let $z = x + iy$. Then $\bar{z} = x - iy$:

$$\begin{aligned} & (x + iy)(x - iy) \\ & + (1 - 2i)(x + iy) + (1 + 2i)(x - iy) \\ & = x^2 + y^2 + x + 2y + i(-2x + y) \\ & \quad + x + 2y + i(2x - y) \\ & = x^2 + y^2 + 2x + 4y \\ & = (x^2 + 2x + 1) + (y^2 + 4y + 4) - 5 \leq 4 \\ & \therefore (x + 1)^2 + (y + 2)^2 \leq 9 \end{aligned}$$

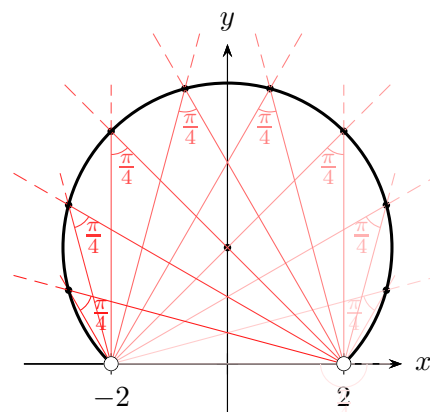
Locus is the disc with centre $(-1, -2)$ and $r = 3$:



(d) i. (2 marks)

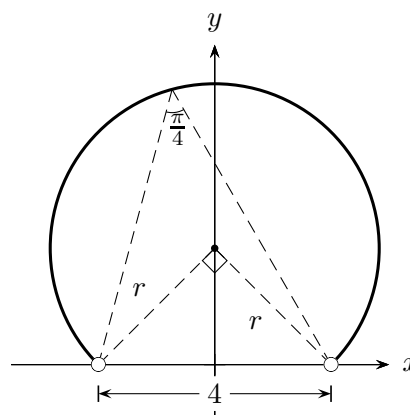
$$\arg(z - 2) - \arg(z + 2) = \frac{\pi}{4}$$

- $\arg(z - 2) = \theta$ results in a straight line, commencing at $x = 2$ where the line makes an angle of θ with the positive x axis (similarly for $\arg(z + 2) = \phi$).
- Angle between the line from $z = 2$ & $z = -2$ respectively is $\frac{\pi}{4}$.



- Hence the locus is a major arc of a circle above the x axis with AB being a chord of the circle, excluding $z = -2$ and $z = 2$.

ii. (2 marks)



- One of these “angles at the circumference” will be made by the diameter of the circle. Hence the angle at the centre of the circle will be $\frac{\pi}{2}$.
- By Pythagoras’ Theorem on the right angled triangle,

$$r^2 + r^2 = 4^2$$

$$2r^2 = 16$$

$$r^2 = 8$$

$$\therefore r = 2\sqrt{2} \quad C(0, 2)$$